CEFAZOLIN

Surgical prophylaxis drug of choice

Cross-allergy

CHEMICAL NAME (6R,7R)-3-{[(5-methyl-1,3,4-thiadiazol-2-yl)sulfanyl]methyl}-8-oxo-7-[2-(1H-tetrazol-1-yl)acetamido]-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (sodium)

CLASS First-generation cephalosporin (parenteral)

TRADE NAMES Ancef · Kefzol · Cefazolin sodium injection

THERAPEUTIC USES / INDICATIONS

- Surgical antibiotic prophylaxis (orthopaedic, cardiac, general, caesarean section).
- MSSA bacteraemia, endocarditis, osteomyelitis.
- Skin and soft-tissue infections; UTI.

PHARMACODYNAMICS

- Inhibits bacterial **cell-wall synthesis** → bactericidal.
- Active against MSSA and streptococci (skin flora); ideal for preventing wound infection.

ADMINISTRATION

- **IV** injection over 3–5 min or infusion; **IM**.
- Surgical prophylaxis: give within 60 min before skin incision; redose if surgery >4 h or major blood loss.
- Use within stability time after reconstitution.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Phlebitis, injection-site pain.
- Diarrhoea, nausea.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis.
- *C. difficile* colitis.
- Seizures at high doses in renal failure.
- Neutropenia, thrombocytopenia (prolonged).

DOSAGES

- **Prophylaxis:** 2 g IV (3 g if ≥120 kg).
- **Treatment:** 1–2 g IV every 8 h (max 12 g/day).
- **Children:** 25–100 mg/kg/day divided every 6–8 h; prophylaxis 30 mg/kg.
- Reduce interval in renal impairment.

CONTRAINDICATIONS & CAUTIONS

- Cephalosporin hypersensitivity; severe penicillin anaphylaxis (use alternative).
- **Caution:** renal impairment.

KEY INTERACTIONS

- **Probenecid:** ↑ levels.
- **Warfarin:** ↑ INR possible.

PHARMACOKINETICS

- **Onset:** immediate (IV). Good bone and tissue penetration; poor CSF penetration.
- Not metabolised.
- Excreted unchanged in urine. **t½:** 1.8 h.

NURSING MANAGEMENT

- Confirm allergy status before theatre.
- Time the dose to finish before incision; document time.
- Monitor for allergic reaction during induction.
- Monitor wound and temperature post-op.
- Check renal function for treatment courses.

PATIENT EDUCATION

- This antibiotic is given to prevent wound infection.
- Report rash, itching or breathing difficulty.

KEY POINTS

- 1st-gen IV cephalosporin — surgical prophylaxis.
- Covers skin flora (MSSA, strep).
- Give within 60 min before incision.
- Redose in long surgery.